Empirical Project : Preparing a Dataset & Replication Package

Introduction to Environmental Economics | AY 2026–27

Work in pairs or groups of 3 for this assignment. For this project, you will prepare a real dataset relevant to an environmental economics question and put together a documented replication package for it. The focus is on building a clean, usable dataset and identifying good questions it could answer.

Concretely, you will: (a) choose or propose a dataset tied to an environmental economics issue; (b) clean and structure it and (c) submit a replication package documenting how you did this, along with 2–3 candidate research questions the dataset could answer.

Choosing a Dataset

Pick a real world dataset and mention the source website/weblink. Only open-access/publicly available datasets are allowed.

Sample Datasets:

Data type	Possible angle	Source
City-level air pollution (India, 2024–25)	Pollution trends, interventions, exposure	https://airquality.cpcb.gov.in/ccr/#/all-india-aqi-portal
Household survey (NSS)	E.g., fuel choice, water access	https://microdata.gov.in/NADA/index.php/catalog/237
Firm-level (ASI)	E.g., electricity prices and power use	https://microdata.gov.in/NADA/index.php/catalog/256
Spatial water-body data	Water resources, climate risk	https://nwdp.nwic.gov.in/dataset/glacial-lakes

Register Your Topic Before You Start

Before you begin cleaning data, add your name(s), chosen dataset, source link, and a title of your intended question(s) to the shared topic-registration sheet. Check the sheet first to avoid duplicating someone else's dataset and question combination – if there is overlap, differentiate by question, geography, time period, or sector.

What Your Replication Package Must Include

Submit a self-contained folder to reproduce your cleaned dataset, with no manual steps. It should contain:

- **README**: dataset source, coverage, and why it is relevant to environmental economics; every cleaning/preparation step, in order.
- **Code**: scripts that take the raw data and produce your cleaned dataset, matching the steps in your README.
- **2–3 candidate research questions**: for each, state the policy relevance and a technique from the course (valuation, instrument comparison, CBA/discounting, treatment-effect reasoning) that could plausibly answer it.

How You'll Be Evaluated

You will be evaluated on (i) the novelty of the topic and research questions proposed with clear links to environmental economics, (ii) coverage of the dataset (time periods, observations), (iii) well-designed correct, clean, and reproducible code to prepare the data for analysis, and (iv) clearly described data cleaning steps in the README file. The source data need to be cited and weblinks mentioned. Use of AI/LLMs is limited to coding/verification. Software choice is entirely up to you – R, Python, Stata, or well-structured Excel/Sheets all work.

The course instructor and Teaching Fellows will review each of your software codes to verify if the data have been cleaned correctly. As LLMs such as Claude can make mistakes, we ask each of to verify each step of your software code to ensure these are correct.

Submission Date: November 30, 2026